

Terahertz Biosensors with high sensitivity for the recognition of neurotransmitter

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Abstract—The quantitative measurement of neurotransmitters with label-free detection methods provides vital clues in tracking neurological functions or disorders. However, there are many challenges for the terahertz biosensor due to the noise and other non-target components like water and so on. Here, we report a rapid quantification method of adrenaline with a specially designed hexagonal open resonant ring. This approach opens opportunities not only for investigating the early diagnostics of neurological disorders or highly sensitive terahertz biosensor applications but also for developing low-cost spectroscopic biosensing applications.

I. INTRODUCTION

THE neurons release neurotransmitters at their synapses which convey information to the neighboring neurons. The secreted neurotransmitters play an essential role in keeping the brain functioning normally. A quite small alteration of the neurotransmitter amount can indicate some brain disorders. Thus, the concentrations of various neurotransmitters serve as candidate biomarkers for the early detection, prognosis, and real-time follow-up of neurological diseases or disorders [1].

Catecholamine is a biogenic compound with many important functions in human body and animals, such as regulating stress, mental activity, mood, learning, sleep, and memory [2]. Catechol includes biological neurotransmitters such as adrenaline (AD), noradrenaline (NAD) and dopamine. The concentration changes of these substances in plasma can lead to changes in pulse rate, respiration, and blood pressure fluctuations. In addition, changes of catecholamine metabolism have been linked to serious disorders of the brain and central nervous system [3-6]. Therefore, the early detection of catechol in the body can diagnose certain diseases and prevent their further development. Another related problem is drug quality control, since treatment with the wrong dose of catecholamines or their synthetic analogues can lead to severe metabolic disorders. Therefore, it is of great significance to develop a sensitive, rapid, economical, and effective method for the determination of catecholamines.

The neurotransmitters will release and clear in the extracellular space very rapidly with low concentration, therefore, the *in vivo* monitoring of the neurotransmitters is so crucial and challenging. Two most widely used techniques for the *in vivo* neurotransmitters monitoring are microdialysis and fast-scan cyclic voltammetry (FSCV), the former is based on a semipermeable probe collecting the analytes which can be analyzed with high performance liquid chromatography (HPLC) externally, which is not real-time measurement strictly and always complex sample preprocessing is prerequisite, the latter is based on the potential difference which is only suitable for partial samples with potential difference [7]. However, these methods still face some technical limitations for clinical uses due to tedious sample preparation, low detection sensitivity,

and poor sample stability. For instance, HPLC require a few minutes to collect sufficient sample volume and currents of the neurotransmitters decay rapidly by hundreds of milliseconds due to diffusion, re-uptake, binding to receptors, or enzymatic breakdown, which may increase the difficulty of sample collection and pretreatment [8]. Then it is urgently required real-time and ultrasensitive detection methods for the molecular identification of neurotransmitters.

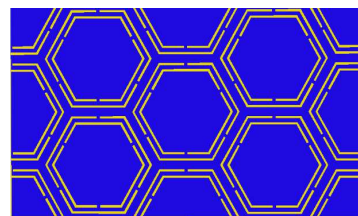


Fig. 1. The geometric configuration of MMs-based biosensor.

Terahertz spectroscopy, being a non-invasive and non-ionizing medical imaging and characterization method, has a great potential for the early diagnosis of tumors or real-time monitoring of disease-related factors. Up to now, many articles are focus on the solid state, which is not in accordance with the actual situation [9] and there is no practical value in medical treatment. However, in addition to the small amount of target analytes in actual biological samples, there are other components such as water. Therefore, it is difficult for the target analytes to be effectively identified. Recent research on surface plasmon resonance (SPR) sensors demonstrate their abilities to achieve local enhancement of the terahertz waves due to resonant effects, pushing the detection limit of existing biosensors [10]. Here, we designed a hexagonal asymmetric split plasmonic ring biosensor as shown in Fig. 1 and demonstrated its potential ability to measure the neurotransmitters.

II. RESULTS

The gold hexagonal pattern was designed on a high-resistivity silicon substrate in CST Microwave Studio. The simulation results show that the hexagonal structure is more sensitive to the imaginary part of dielectric constants, which is quite consistent with our previous experiments that the imaginary part of the dielectric constant changes more obviously than the real part at different concentrations in the terahertz frequencies. The hexagonal pattern density and the corner sharpness of the hexagonal shape were also investigated by the simulation to optimize the performance of the sensor. Finally, this terahertz biosensor was applied to a rapid quantitative analysis of the adrenaline. The reflection coefficient of the sensor loaded with different concentrations of the adrenaline aqueous solutions was measured with the

reflection module of a commercial terahertz time domain spectroscopy system (TeraPulse 4000, TeraView Ltd.). The result is shown in Fig. 2 (a) and (b). We can see that there is a good linear relationship between the reflection coefficients and the concentrations of the adrenaline.

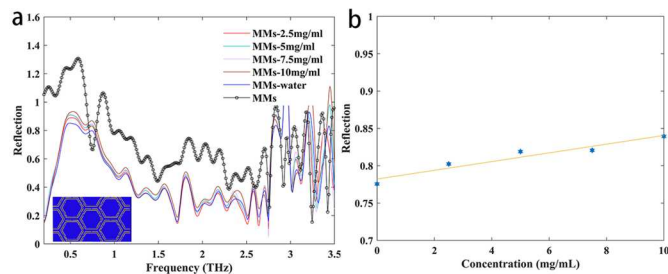


Fig. 2. (a) Reflection coefficients of different concentrations of adrenaline solutions dropped on the terahertz biosensors from 0.2 to 3.5 THz (b) the reflection coefficients plot with the concentrations at 0.67 THz.

To demonstrate the improved sensitivity of our device, a series of dilute solutions with different concentrations were prepared in the range of 0-11 mg/mL and measured on quartz as well as on our device for comparison. Fig. 3 shows the results tested on the quartz. From Fig. 3(a), it can be seen that the reflection coefficients almost overlapped with each other. Then the data was analyzed by the iPLS method with iToolbox, a free software developed by Lars Nørgaard and Riccardo Leardi [11]. To get rid of redundant information and build optimal models, it is necessary to use the variable selection. The interval numbers ranging from 5 to 30 were tested, and the iPLS calibration can get the best RMSEP and R when intervals set as 9 with the frequency of 1.42–1.58 THz. Fig. 3(b) shows the optimal spectral intervals selected by iPLS. In this region, the mean RMSECV can reach 1.93 ± 0.53 and the correction coefficient of adrenaline (R) is 0.7996.

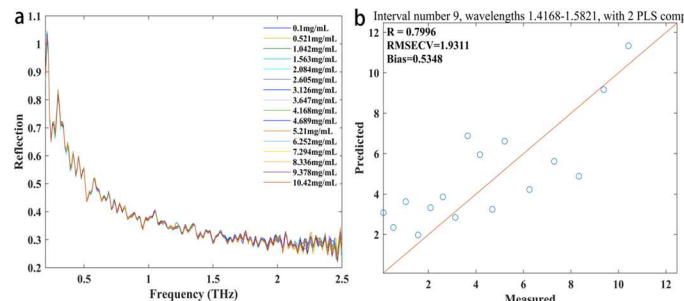


Fig. 3. (a) Reflection coefficients of different concentrations of adrenaline solutions dropped on the quartz from 0.2 to 2.5 THz (b) Correlation coefficient of the iPLS predicted content versus the actual content of the adrenaline.

Fig. 4 shows the results tested on our terahertz biosensors. It is shown in Fig. 4(a) that the reflection coefficients have a much clearer variation for different concentrations compared to that plot in Fig 3(a). Similarly, the data was analyzed with the iPLS method with iToolbox. The interval numbers ranging from 5 to 30 were tested, and the iPLS calibration can get the best RMSEP and R when intervals set as 8 with a frequency range of 1.24–1.4 THz. Fig. 4(b) shows the optimal spectral intervals selected by iPLS. In this region, the mean RMSECV is 1.27 ± 0.07 and R is 0.9228, which has a big improvement compared to those results tested on quartz.

To conclude, our terahertz biosensors can improve the sensitivities and provide a more credible and accurate result

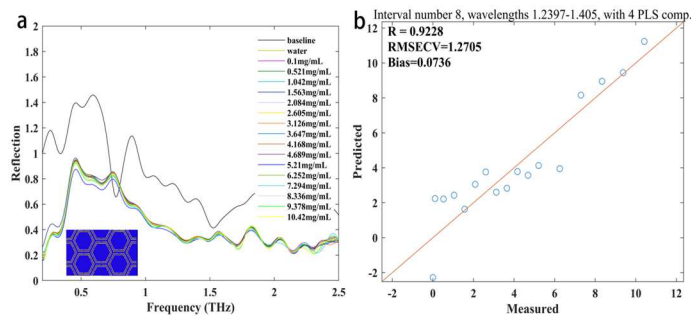


Fig. 4. (a) Reflection coefficients of different concentrations of adrenaline solutions dropped on the terahertz biosensors from 0.2 to 2.5 THz (b) Correlation coefficient of the iPLS predicted content versus the actual content of the adrenaline.

than conventional sampling windows. It may provide a new idea for the detection of dilute biological solutions. In future work, real sample environments such as plasma, mineral salts and other interference factors need to be considered.

III. ACKNOWLEDGEMENT

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